Paleomagnetism and orogenic history of the Botwood Group and Mount Peyton Batholith, Central Mobile Belt, Newfoundland[1]

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This is the first paleomagnetic result from the Paleozoic Botwood Central Zone of Newfoundland. Two Lower Paleozoic rock units were sampled: the Botwood Group (rhyolite and red sandstone) at 16 sites and the Mount Peyton Batholith (dioritic and granitic phases) at 12 sites. Two phases are recognized in this intrusion: dioritic (420 Ma) and granitic (380 Ma). Four distinct magnetization directions were observed during the study. In the Lawrenceton Formation (Botwood Group, rhyolite) the direction obtained after tilt correction (180°, +47°) gives a pole at 125°E, 13°N, probably of Lower Silurian age. The overlying Wig Wam Formation of the Botwood Group (red sandstone) was magnetized during a reversal. Detailed demagnetization techniques (thermal, chemical, and alternating field) were used to separate the normal and reversed directions in composite magnetizations; with a mean direction of 194°, +28°, the pole (100°E, 25°N) obtained after tilt correction is related to a Silurian age. Two magnetizations were found in the Mount Peyton Batholith: the first one (pole at 68°E, 15°S) is associated with the dioritic phase with an age of 420 Ma. The second one (pole at 125°E, 63°S) is associated with the granitic phase at 380 Ma.

Cette étude représente le premier résultat paléomagnétique provenant de la zone centrale paléozoïque de Terre-Neuve (zone de Botwood). Deux types de roche du paléozoïque inférieur ont été échantillonnés: le groupe de Botwood (16 sites) et l'intrusion du Mont Peyton (12 sites). L'intrusion est composée de deux phases: dioritique (420 Ma) et granitique (380 Ma). L'étude paléomagnétique de ces types de roche a révélé la présence de quatre aimantations distinctes. Dans le groupe de Botwood, la formation Lawrenceton (rhyolite) contient une aimantation stable, qui, après correction pour le pendage (180°, + 47°), donne un pôle (125°E, 13°N) probablement du Silurien inférieur. La formation de Wig Wam qui repose sur la formation de Laurenceton est composée principalement de grès rouges. Cette formation recelait deux directions anti-parallèles résultant d'un renversement du champ magnétique à cette époque. Après correction pour le pendage, on obtient une grande moyenne de (194°, +28°) avec un pôle (100°E, 25°N) probablement d'âge Silurien (moyen à supérieur) pour ces grès rouges. Deux aimantations très différentes ont été obtenues de l'intrusion du Mount Peyton. La première est associée à la phase dioritique (420 Ma) avec un pôle à 068°E, 15°S. La seconde est associée à la phase granitique (380 Ma) avec un pôle à 125°E, 63°S.

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Introduction

The island of Newfoundland can be subdivided into eight tectonic–stratigraphic zones: Lomond, Hampden, Fleur de Lys, Notre Dame, Exploits, Botwood, Gander, and Avalon (Fig. 1; Williams *et al.* 1974; Williams and Stevens 1974). The three central zones (Notre Dame, Exploits, and Botwood) are defined as oceanic crust (in part), overlain by volcanic and sedimentary rocks of Lower Paleozoic age. This presentation deals exclusively with rocks of the Botwood zone, and more specifically the Botwood Group and the Mount Peyton Batholith (Fig. 2).

The Botwood zone consists of Ordovician slate overlain by Silurian conglomerates, terrestrial volcanics, and red fluviatile sandstones (Twenhofel and Schrock 1937; Williams 1962).

The Botwood Group and the Mount Peyton Batholith were sampled for a paleomagnetic study. The Botwood Group itself is composed of basal red conglomerate, greywacke, and shale, overlain by lava, tuff, and agglomerate (Lawrenceton Formation), in turn covered by red sandstone, siltstone, and conglomerate (Wig Wam Formation) (Dean 1976). In the area studied (Fig. 2), the Botwood Group is cut by a two-phase (dioritic and granitic) intrusion; the Mount Peyton Batholith.

Geology

Botwood Group

In the Grand Falls – Bishop Falls area (Fig. 2), the Botwood Group is represented mainly by purple and red rhyolite (Lawrenceton Formation), overlain by red sandstone (Wig Wam Formation; Dean 1976). In the town of Grand Falls the contact between the rhyolite and the sandstone is sharp and

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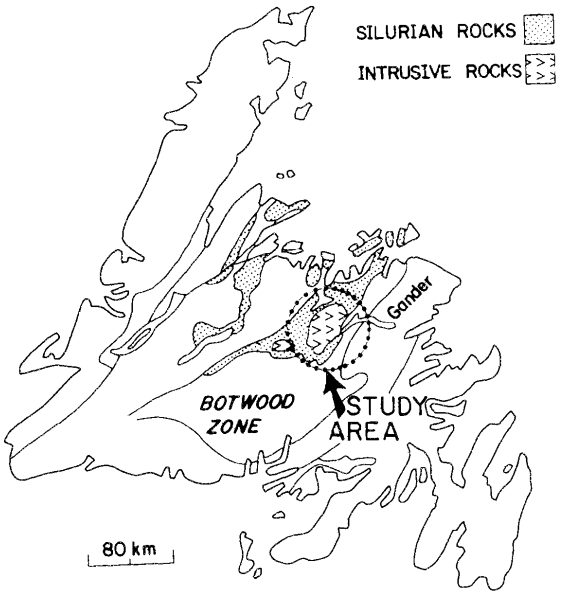


FIG. 1. Map of Newfoundland, showing the area studied in the Botwood tectono-stratigraphic zone.

little intermingling of the two is seen. The sandstone is red to brown, fine-grained micaceous, and always shows cross-laminations. Ripple marks and cross-bedding indicate a shallow water environment (Malpas and Kennedy 1974). The beds form a wide northeast trending belt covering most of the Botwood zone; the strikes are generally parallel to the belt and the dip rises from 45–90°. Most of the rocks in the area have been affected by low regional metamorphism (greenschist facies), but contact metamorphic aureoles in the Botwood Group around the Mount Peyton Batholith have been detected (Williams 1962).

Mount Peyton Batholith

The Mount Peyton Batholith is a bimodal intrusion about 60 km long by 30 km wide in central Newfoundland (Fig. 2). The intrusion is composite, the first phase is dioritic in composition, the second is granitic to monzonitic. At the edge of the intrusion a mixture of the two phases can be seen in the form of blocks of diorite floating in the granitic matrix. A comprehensive description of the petrol-

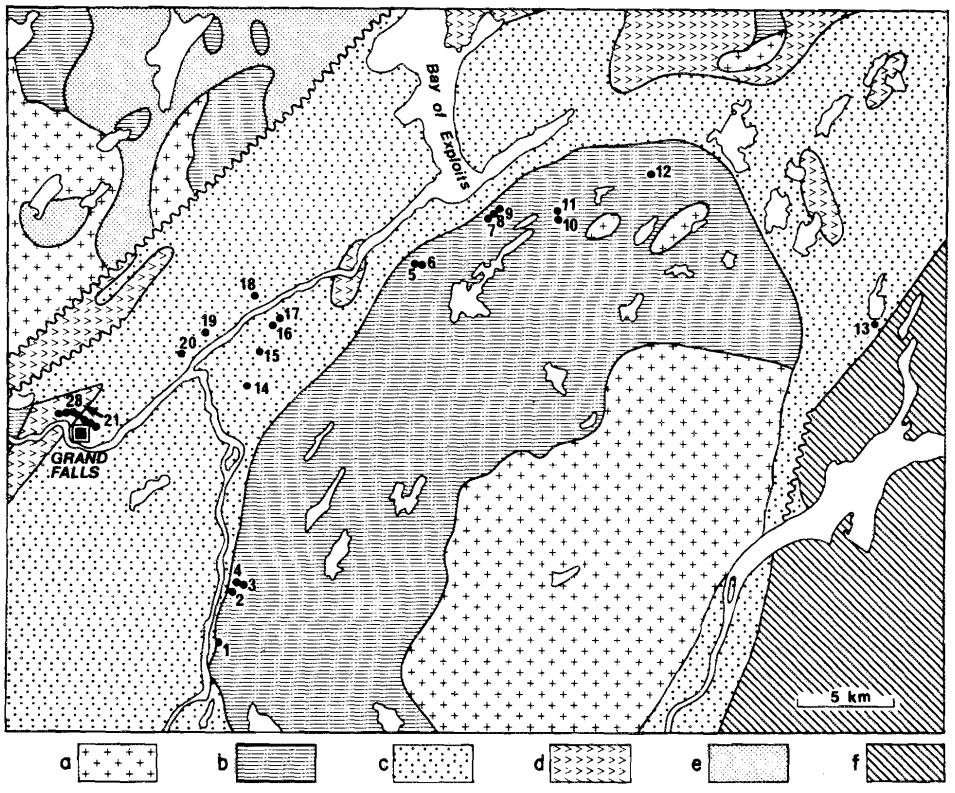


FIG. 2. Geological map of the Mount Peyton area, Grand Falls, Newfoundland, showing the sampling site localities. The different geological units are as follows (*a*) granite (late phase); (*b*) diorite (Mount Peyton early phase); (*c*) Botwood Group, Wig Wam Formation; (*d*) Botwood Group, Lawrenceton Formation; (*e*) Wild Bights Group; and (*f*) Gander Group.

TABLE 1. Radiometric data and paleomagnetic sampling sites

Rock unit	Rock type	Radiometric method (Ma) K/Ar[a]	Rb/Sr[b]	Paleomagnetic sites
Mount Peyton Batholith	Granite	380 ± 50 264 ± 52	375 ± 15	7
	Diorite	410 ± 21		5
	Intrusive contact			
	Wig Wam Formation (Red sandstone)	423, 422 ± 17[c]		13
Botwood Group	Lawrenceton Fm. (Rhyolite, lava)			3

[a]K/Ar ages are from Williams (1964).
[b]Rb/Sr ages are from Bell and Blenkinsop (1975) using 1.47×10^{-11} year^{-1} (397 Ma using 1.39).
[c]These ages are considered to have been reset by the diorite phase of the Mount Peyton intrusion (Williams 1964).

ogy and geochemistry is found in Strong (1978), together with a more complete geological reference list.

Age Relationship

Based on geological evidence, Williams (1964) and Williams *et al.* (1974) recognized that two orogenic events affected the area studied during the Paleozoic; the first in middle to late Ordovician time is tentatively associated with the Taconic orogeny. The second, which is well defined by tight folding, metamorphism, and igneous intrusions, is probably post-middle Silurian in age and related to the Acadian orogeny (Devonian).

Isotopic ages in the studied area are shown in Table 1. Williams (1964) concluded that on the basis of radiometric data, the intrusive activity was not confined to a single geological epoch.

Sampling and Techniques

The sampling consisted of 197 cores (2.5 cm in diameter) obtained from 28 sites (16 from the Botwood Group and 12 from the Mount Peyton Batholith). From a core, one, two, or three specimens (height 2.2 cm) were obtained and a total of 352 specimens were studied. Three methods of cleaning were used: alternating field (AF), thermal, and chemical. Thermal and AF demagnetizations were carried out using apparatuses operating in residual fields 2 nT (Roy *et al.* 1969, 1972, 1973) and a Schonstedt thermal unit (TSM-1). All measurements were performed using a Digital Schonstedt spinner magnetometer (DSM-1).

Thermal Demagnetization

All thermally treated specimens were demagnetized in steps of 100°C to 500°C and thereafter 550, 560, 570, 580, 600, 625, 650, and 670°C steps.

Chemical Leaching

The treatment consisted of immersing the specimen (after drilling a 0.6 cm axial hole to increase the exposed surface; Park 1970) in hydrochloric acid for the following accumulated times: 25, 100, 250, 550, 950, 1300, and 2300 h—washing in water, storing in zero field for two days, measuring, and reimmersing in fresh acid. Hydrochloric acid (10 *N*) was used throughout the treatment owing to low permeability of the rocks.

AF Demagnetization

All specimens treated were subjected to fields of 5, 10, 15, 20, 25, 30, 40, 50, 75, 100, and 125 mT. Some specimens were subjected to fields up to 290 mT, but no appreciable changes between 125 and 290 mT were detected.

Results

From the rock units studied, four different magnetizations were found (grouping A–D). They are discussed from the oldest (A) to the youngest (D).

Grouping A

Grouping A magnetization is found in the Lawrenceton Formation (purple rhyolite) of the Botwood Group. Three sites were sampled and a total of 18 cores were analyzed. Their results are listed in Table 2. Natural remanent magnetizations (NRM) are plotted with their related site means in Fig. 3*a*. Site 26 is the only site with a good within-site grouping. The poor grouping at sites 27 and 28 is evidenced by the low precision index (k) (Table 2). After cleaning, both within and between site dispersion are greatly reduced (much larger k) (Figs. 3*b*, *c*). Evidently, all specimen directions belong to the same population. It is interesting to note the difference in statistical parameters when one uses the sample or the site as unit weight; the use of sample unit weight yields a much smaller radius of circle of confidence (α_{95}). However, the site unit weight method is considered more representative and is used throughout this paper.

Both AF and thermal demagnetizations were

TABLE 2. Site mean direction of magnetization and pole position for Lawrenceton Formation (rhyolite) of the Botwood Group

	NRM					Demagnetization*					Site pole		
Site	N	D	I	R	k	N	D	I	R	k	Lat. (S)	Long. (E)	α_{95}
26	7	254	+27	6.99	485	7	254	+28	6.97	186	01	056	4.4
27	7	310	+04	4.22	02	7	258	+38	6.95	130	08	057	5.0
28	4	237	+64	2.81	03	4	261	+30	3.98	149	07	052	7.5
Mean						3	258	+32	2.988	177	05	056	9.2
Tilt corrected							180	+47			13N	125E	

*Demagnetization in field of 25 mT or temperature of 570°C.
NOTES: N = number of specimens; R = resultant vector; k = precision estimate (Fisher 1953); and α_{95} = radius of circle of confidence (P = 0.05). Natural remanent magnetization (NRM) intensities vary from 100 to 500 × 10^{-3} A m^{-1}.

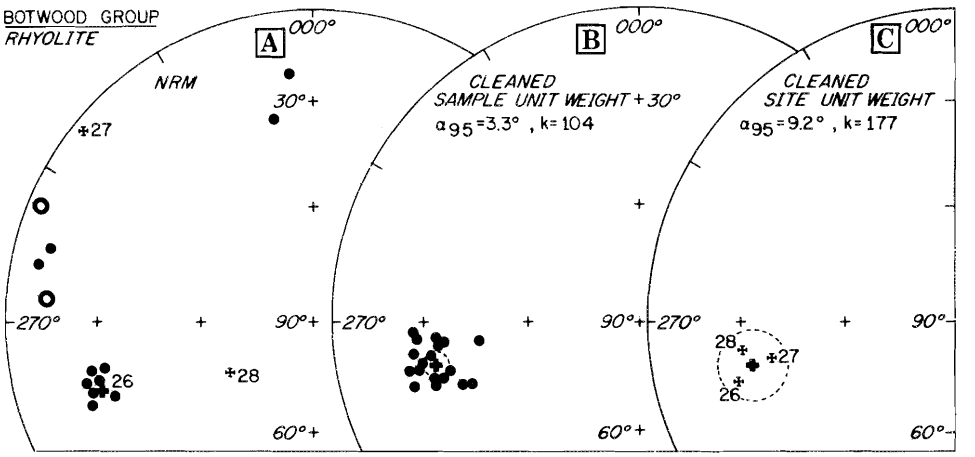


FIG. 3. (*a*) Distribution of NRM directions from the Lawrenceton Formation rhyolite; full (open) symbols are down (up) inclinations, crosses are site mean. (*b*) Distribution of directions after thermal and AF cleaning (sample unit weight). (*c*) Distribution of site means. All directions plotted before tilt correction.

applied. Demagnetization by AF was performed in steps of 2.5 mT up to 10 mT and thereafter in steps of 10 mT up to 100 mT. Although generally stable up to 100 mT, the best grouping is obtained by cleaning in a field of about 25 mT (Fig. 3*b*). An example of an AF demagnetization path is shown in Figs. 4*a* and 4*b*; a cleaning field of 7.5–10 mT takes the direction to its end point position. Thermal demagnetization was performed in steps of 100°C to 500°C and subsequently at 550, 570, and 575°C. The direction reaches an end point in the 400–500°C range and remains in that position until 575°C. In general, the end point is more easily defined by the AF than by the thermal technique.

The three sites of this grouping were sampled at the contact of the rhyolite and sanstone. The sediments in contact with the rhyolite have been folded (031°, + 70°) and the rhyolite outcrop seems to have followed that tilt. The rhyolite poles, before and after tilt correction, are 056°E, 05°S and 125°E, 13°N respectively.

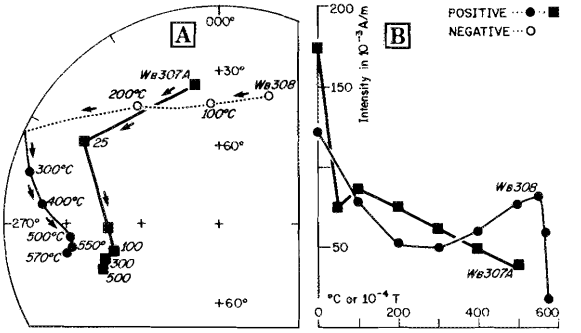


FIG. 4. Magnetization response to thermal (circle) and AF (square) demagnetization: (*a*) direction; (*b*) intensity decay curve. All directions plotted before tilt correction.

Grouping B

The Wig Wam Formation red sandstone unit overlies the Lawrenceton Formation; 92 cores were sampled at 13 sites. These red beds acquired their remanence at a time of field reversal. The two polarities of the field are clearly evident at some of

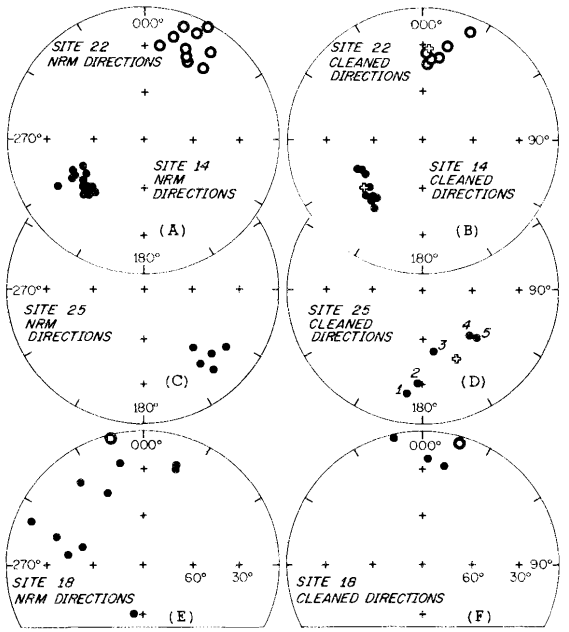


FIG. 5. Parts *a*, *c*, and *e* represent NRM directions from the Wig Wam Formation; *b*, *d*, and *f* their related cleaned equivalents. Full (open) symbols represent positive (negative) inclinations. Crosses are site mean directions. (*d*) The numbers 1–5 are discussed in the text. Cleaned symbols stand for AF, thermal and chemical undifferentiated. All directions plotted before tilt corrections.

the sites (Figs. 5*a*, *b*). At other sites the directions are scattered, indicating that in the present case the magnetization was probably acquired during a field reversal. It will be shown later that these (intermediate) directions actually represent the vector sum of two magnetizations acquired, respectively, before and after the field reversal.

It has been shown that the initial magnetization of red beds is a lengthy process spanning field reversal time scales and which can sometimes be divided into at least three phases (Roy and Park 1974). Chemical, thermal, and high AF treatments can separate the late phase from the earlier phases of magnetization (Roy and Lapointe 1976). However, only chemical leaching could separate all three phases of magnetization (a detrital (DRM) and two chemical (CRM_A and CRM_B) remanent magnetizations) in Carboniferous red beds (Roy and Park 1974).

Chemical, thermal, and AF cleaning techniques were performed on the Wig Wam red beds. Unfortunately, the different phases of magnetization could not be separated by chemical leaching. The effectiveness of the chemical method is highly dependent upon the penetration rate of the acid which is regulated by the porosity and permeability of the rock. Ideally, the rate should be rapid so that the grains carrying the late phase magnetization (red pigmentation CRM_B) can be destroyed throughout the specimen before the more resistant (to acid) grains carrying the early phases (DRM and CRM_A) are attacked. The Botwood samples are fine-grained sediments and acid penetration is very slow. The slicing of specimens after 2300 h in HCl shows that 15% of the red cement remains in the sample which may mean that in the outer zone all phases are attacked to varying degrees, while in the inner zone none of the phases are destroyed.

Without the benefit of chemical leaching none of the phases can be isolated experimentally and this limits the number of field direction determinations that one can obtain. This is the reason why $N = 72$ in Table 3, although 92 cores were sampled. The cores were sliced to produce one, two, or three specimens per core. Specimens from the same core would be subjected sometimes to different treatments, sometimes to the same treatment. Magnetic carriers react differently to different treatments so that in general the specimen direction is used as unit weight. When two or three specimens are subjected to the same treatment, the cleaned core mean direction is used as unit weight. For example, at sites 14 and 22 (Fig. 5) the NRM directions shown are the specimen directions (Fig. 5*a*) and the cleaned directions are the core mean directions (Fig. 5*b*).

The possibility of obtaining a field direction determination depends on the timing of the field reversal with respect to the magnetization process. If the field reversal occurred at the very beginning of (or before) the magnetization process or toward the end of (or after) the magnetization process, then all phases would have the same polarity. This seems to have happened at sites 14 and 22 (Fig. 5*b*) with the fairly tight grouping and opposite polarities. The two groups are not exactly antiparallel (143°). This may be owing to secular variation or to the fact that the reversal occurred after the beginning and (or) before the end of the magnetization process. The stability of this magnetization is shown in Figs. 6 and 7. In the Carboniferous red beds studied by Roy and Park (1974), the late phase magnetization carried by the red cement could be removed by thermal demagnetization between 550 and 670°C. If this is so for these beds, then the red cement contributes very little (~ 10%) to the total magnetization. The fact that upon cleaning up to 600°C the intensity increases in the northerly directed grouping (Fig. 7), indicates that the reversal occurred at the end of the magnetization of these sites, and that the northern polarity predates the southern polarity. This is in accordance with the site stratigraphic

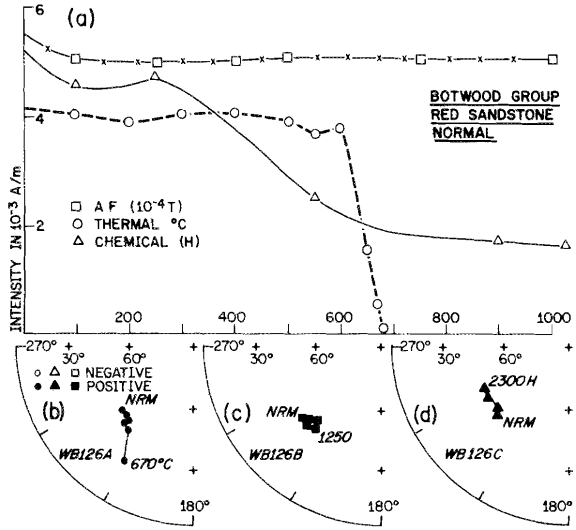


FIG. 6. (*a–d*) Response of specimens to different cleaning techniques. (*a*) Intensity decay curves; (*b*) thermal treatment; (*c*) alternating field (AF); and (*d*) chemical leaching. All directions plotted after tilt corrections.

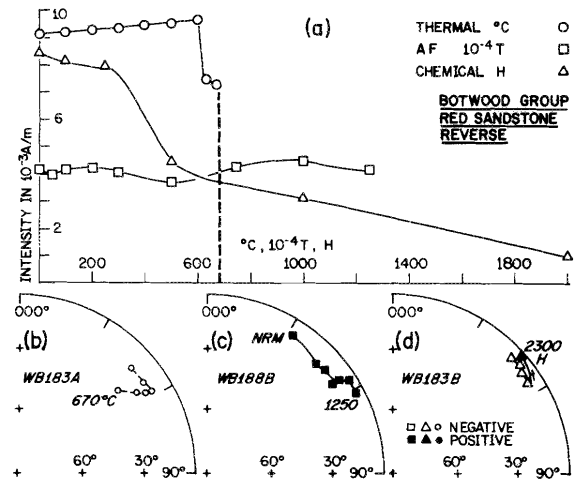


FIG. 7. (*a–d*) Response of magnetization to different cleaning techniques. (*a*) Intensity decay curves; (*b*) thermal treatment; (*c*) alternating field treatment; and (*d*) chemical leaching treatment. All directions plotted after tilt correction.

levels; the northern polarity is found at the lower part of the section (Fig. 2, Site 21-25).

During cleaning, the intermediate directions sometimes are displaced toward one of the mean directions of sites 14 or 22 (Fig. 8); sometimes they move to another intermediate direction (Fig. 9). This is attributed to overlapping characteristic magnetic spectra of oppositely directed magnetizations (Roy and Robertson 1968). Slicing of specimens along the bedding plane (Fig. 9) shows that the direction of the whole specimen was indeed

TABLE 3. Normal and reverse directions of magnetization in Wig Wam Formation (red sandstone)

	Cleaning treatment									Sites mean value						Pole				
	Thermal			AF			Chemical			Tilt uncorrected			Tilt corrected			Tilt Uncorrected		Tilt corrected		
Site	*N*	*D*	*I*	*N*	*D*	*I*	*N*	*D*	*I*	*N*	*D*	*I*	*D*	*I*	*N*	Lat.	Long. (East)	Lat.	Long. (East)	*k*	α_{95}
13	7	022	+36	4	031	+47	3	041	+43	14	021	+41	359	−15	14	59N	084	34N	126	17	9.8
14	4	179	+12	3	179	+13	2	176	+26	9	179	+15	229	+45	9	33N	126	03N	082	88	5.5
16	4	166	+07	3	173	+11	2	162	+28	9	162	+15	224	+48	9	32N	146	03N	087	06	23.3
17	5	152	+09	2	145	+13	2	147	+40	9	145	+19	237	+63	9	24N	153	16S	087	05	24.3
15	3	195	+18	2	197	+20	2	183	+34	7	192	+24	221	+37	7	28N	111	12N	086	47	8.8
18	3	013	+39	0	—	—	1	002	+45	4	017	+46	179	+09	4	61N	089	46N	126	29	17.4
19	2	008	+25	1	349	−04	2	007	+42	5	004	+26	003	−24	5	55N	119	29N	121	10	25.4
20	2	037	+38	0	—	—	0	—	—	2	037	+38	022	−03	2	49N	064	36N	097	—	—
21	2	038	−30	2	040	−20	1	037	−21	5	039	−24	009	−13	5	20N	084	34N	114	47	11.3
22	3	057	−33	3	060	−32	0	—	—	6	059	−32	002	−31	6	06N	070	24N	123	15	17.5
23	1	223	+38	3	205	+40	0	—	—	4	210	+39	173	+09	4	14N	096	36N	133	9	32.8
24	1	055	−33	0	—	—	0	—	—	1	055	−33	001	−29	1	—	—	—	—	—	—
25	2	244	+40	1	270	+43	1	—	—	3	238	+38	184	+36	3	03N	072	21N	120	—	
Mean	13	202	+08	10	197	+19	8	184	+04	13	200	+09	194	+28	13	34N	100E	25N	110E		

NOTES: $R = 9.89$; $k = 3.8$; $\alpha_{95} = 24.3$. $R = 11.71$; $k = 9.3$; $\alpha_{95} = 14.3$. *NRM intensities vary from 5 to 50 × 10^{-5} A m^{-1}.

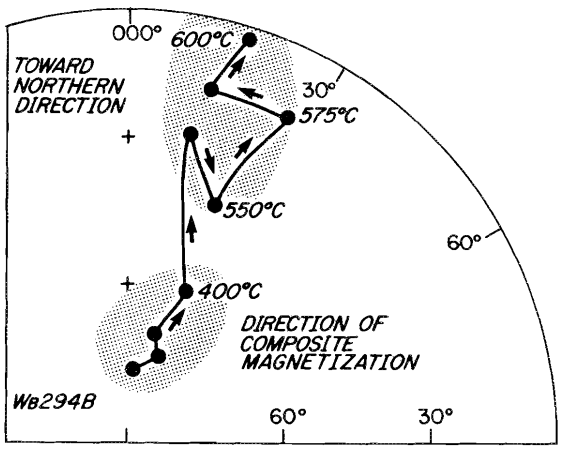


FIG. 8. Response of magnetization to thermal cleaning. The NRM direction belongs to composite magnetization, but it slowly moves toward the reversed direction, without reaching it. Full symbols represent downward inclinations. All directions plotted before tilt correction.

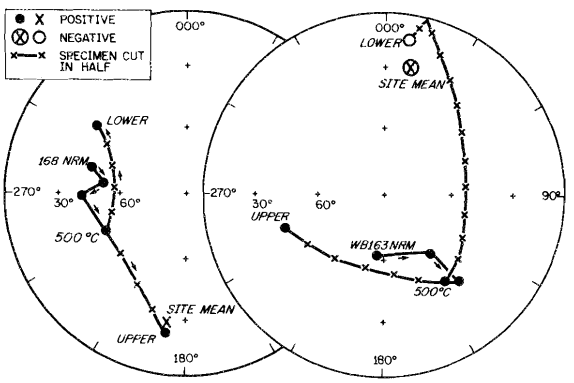


FIG. 9. Response of magnetization with intermediate direction during thermal cleaning. After 500°C the specimens are cut in half (for more details see Roy and Park (1974) and Roy and Lapointe (1978)) along the bedding plane and remeasured. Thereafter they are treated like a complete specimen, reheated and remeasured. The halves clearly show that a field reversal occurred during magnetization. Both halves have a tendency to move towards the reversed and normal Wig Wam direction. All directions plotted befor tilt correction.

the direction of the resultant vector of two magnetizations, one of which having a direction near the mean direction (Roy and Lapointe 1978).

It then appears that for some specimens, the reversal occurred during the late phase and for other specimens it occurred during the early phases, which cannot be separated. It is therefore difficult to establish criteria for acceptance or rejection of data.

Selection of data (Table 3) is based mainly on the behaviour of the directions during cleaning (especially the direction after the last cleaning) and partly on the subjective judgement of the author. For example, at site 25 (Fig. 5*d*) directions 1, 2, and 3 were accepted because upon cleaning they moved in the general direction of the mean. The directions 4 and 5 were rejected because they oscillated about the original direction. At site 18 (Fig. 5), four directions definitely moved toward the mean while the other directions behaved in a random fashion. It is noted that the site mean direction is northerly directed at six sites and southerly directed at seven sites. This may well cancel the effects of improper resolution of the different phases of magnetization and hence the mean (six sites northerly directed were reversed and averaged with the seven sites southerly directed) should be a fairly good estimate of the field mean direction at the time of magnetization.

Directions in all figures are given after correction for tilt, assuming that the magnetization is pretilting. This assumption is reasonable for three reasons: (*a*) the cleaning experiments indicate that the magnetization was acquired during the initial process of magnetization, (*b*) the k (9.3) after correction for tilting is larger than the k (3.8) before correction, and (*c*) both the preceding rhyolite mean direction (180°, +47°) and the following diorite mean direction (254°, +53°) are in better agreement with the after correction mean direction (194°, +28°) than with the before correction mean direction (200°, +09°).

Groupings C and D

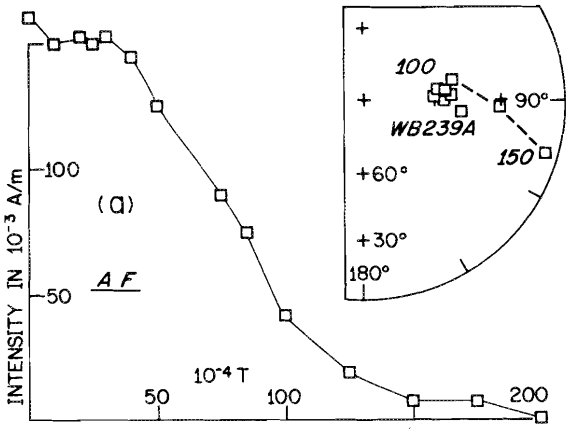
The two phases of the Mount Peyton Batholith (Fig. 2) were sampled; five sites (42 cores) from the dioritic phase (grouping C) and seven sites (67 cores) from the granitic phase (grouping D). However, as mentioned earlier, contacts between the dioritic and granitic phases take the form of a mixture of both units: blocks of diorite floating into granitic matrix. These blocks, together with the granitic matrix, were sampled at two sites (5 and 6).

Thermal and AF cleaning treatments show that the diorite is carrying a very hard magnetization (Fig. 10). The blocking temperature spectrum of that magnetization is limited to a narrow window (500–570°C) and the remanent coercive force spectrum is from 40 up to 100 mT. Both treatments remove another less stable magnetization which has distributed blocking temperature below 200°C and remanent coercive forces below 20 mT. The direction of this superimposed magnetization is randomly distributed about the site mean direction, except at site 6 where it is southerly directed. The removal of this parasitic magnetization is well illustrated by the large increase in k upon cleaning (Table 4).

TABLE 4. Sites mean value of Mount Peyton intrusion 'First phase, diorite'

Site	NRM N	NRM D	NRM I	Demagnetized k	Field (mT)	Temp. (°C)	N	D	I	R	k	Pole Lat. (S)	Pole Long. (E)	α_{95}
21	8	118	−48	13	50	550	8	078	−62	7.8333	42	25	073	09
20	7	107	−48	76	50	500	7	081	−59	6.9879	497	24	069	03
19	7	086	−41	02	20	400	7	085	−54	6.7173	21	22	062	13
18	11	059	−27	06	30	400	7	070	−43	6.9739	230	06	066	04
17	9	178	−13	07	50	200	5	062	−44	4.9459	74	03	072	90
Site mean	5	109	−43	04			5	074	−53	4.9345	61	15	068	9.8

NOTE: NRM intensities from 100 to 400 × 10^{-3} A m^{-1}.



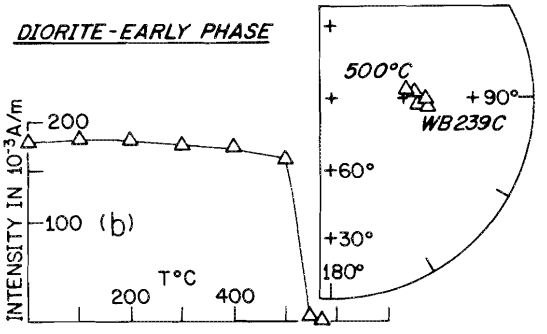


FIG. 10. Response of magnetization of two specimens from the same core to AF cleaning. (*a*) Stability is up to 100 × 10^{-4} T, but after 150 × 10^{-4} T the direction became meaningless. (*b*) Stability up to 500°C; at 550 and 570°C the directions are meaningless and not shown.

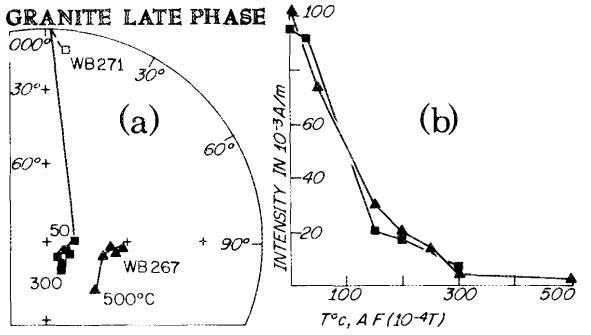


FIG. 11. (*a*, *b*) Response of magnetization in samples from the late granite phase of Mount Peyton. Stability on AF and thermal treatments is up to 300 × 10^{-4} T and 500°C respectively.

The most stable magnetization carried by the granite has distributed blocking temperature up to 500°C and remanent coercive forces up to 30 mT (Fig. 11). A superimposed magnetization is easily removed by cleaning in fields of 5–10 mT or heating to 100°C. Obtention of the stable magnetization by AF requires a very slow decrease of the field (peak to zero in 4 min). The mean site direction (001°, +83°; Table 6) is very different from that obtained from the diorite (074°, −53°; Table 4), suggesting different ages for the two phases.

Results from sites 5 and 6 indicate that the magnetic characteristics of both diorite and the granitic matrix are similar to those of the granite rocks (grouping D). A comparison of mean directions (Watson 1956) shows that they belong to the same population (Table 5). The combined mean direction is 038°, +85°. The agreement between this direction and that of the granite indicates: (*a*) that they were magnetized at the time of the granitic phase and (*b*) the granitic intrusion postdates the dioritic intrusion. This is in accordance with radiometric and geochemical evidence reported by Williams (1964), Bell and Blenkinsop (1975), and Strong (1978). The grouping C pole (068°E, 15°S) is then associated with the diorite age (420 Ma) and the grouping D pole (125°E, 63°S) (Table 6) to the granite age (380 Ma). It is to be stressed that although the grouping D pole is proposed to be related to an age of 380 Ma, it is possible that this magnetization represents a younger event. Since no contact tests were applicable to this rock unit, further work will be needed before this idea can be more conclusive.

TABLE 5. Comparison of direction between the diorite inclusions and the granitic matrix

Rock unit	N*	D	I	Pole Lat. (S)	Pole Long. (E)	R	k	α_{95}
Diorite inclusions	23	077	+85	50	139	9.33	14	13.6
Granite matrix	14	024	+83	61	136	13.41	22	8.6
Combined†	24	038	+85	56	136	22.70	18	7.2

*N = number of specimens; all statistics are based on specimen unit weight.
†Watson's (1956) test gives a positive test ($0.698 < 3.20/_{0,05}$) from F-ratio.

TABLE 6. Site mean direction of magnetization and site pole for Mount Peyton intrusive granitic (second phase)

Site	NRM* N	NRM* D	NRM* I	NRM* k	Cleaned N	Cleaned D	Cleaned I	Cleaned R	Cleaned k	Site pole Lat. (S)	Site pole Long. (E)	Site pole α_{95}
1	11	065	+80	17	11	014	+81	10.86	72	66	135	5.0
2	11	359	+84	96	11	357	+81	10.81	54	66	122	6.3
3	10	018	+85	86	7	004	+84	6.65	17	60	126	14.9
4	7	328	+69	30	5	293	+57	4.71	14	42	149	21.2
5	11	033	+54	05	9	084	+74	8.72	29	44	167	10.4
6	9	339	−08	02	9	070	+78	8.80	41	51	160	08.3
12†	8	353	+26	03	—	—	—	—	—	—	—	—
Mean sites value					6	001	+83	5.79	24	63S	125E	13.8

*Site 12 does not appear to carry a stable magnetization and was rejected.
†NRM intensities vary from 50 to 300 × 10^{-3} A m^{-1}.

TABLE 7. List of paleomagnetic poles from Newfoundland

Rock unit	Pole Lat.	Pole Long.	Index	Rock unit age	Authors
Cloud Mountain	05N	172E	1A	Lower Cambrian	Rao and Deutsch (1976*b*)
Bradore Fm.	09N	149E	1B	Lower Cambrian	Black (1964)
Bradore Fm.	29N	167E	1C	Lower Cambrian	Rao and Deutsch (1976*a*)
Steel Mountain	22N	139E	2A	451 Ma	Murthy and Rao (1976)
Betts Cove	28N	158E	2B	Ordovician	*In* Morris (1976)
St. George Ls.	26N	126E	2C	Ordovician	Beales *et al.* (1974)
Moreton's Harbour	32N	129E	2D	Ordovician	*In* Morris (1976)
Wabana Gr.	28N	168W	2E	Ordovician	Deutsch and Rao (1970)
St. George Gr.	30N	119E	2F	Ordovician	Deutsch and Rao (1976)
Botwood Gr. (rhyolite)	13N	125N	3A	Silurian	This paper
Botwood Gr. (sandstone)	25N	113E	3B	Silurian	This paper
Mount Peyton (diorite)	15S	068E	3C	420 Ma	This paper
Mount Peyton (granite)	63S	125E	3D	380 Ma	This paper
Clam Bank Gr.	28N	139E	4A	Devonian	Black (1964)

Synopsis of Results

Silurian and Devonian rocks from the Botwood zone of Newfoundland yield four distinct magnetizations (groupings A–D), each one associated with a different geological unit. According to geological and radiometric evidence, the Botwood Group, composed of volcanics (grouping A) and sedimentary (grouping B) rocks, has been subjected to folding; it is intruded by a two-phase (groupings C and D) body, the Mount Peyton Batholith. The most reasonable interpretation is that the A and B magnetizations were acquired prior to folding, and they are given a Lower Silurian age. They yield poles at 125°E, 13°N and 110°E, 25°N respectively. A comparison of directions between diorite inclusions and their granitic matrix shows that the C magnetization, found in the diorite phase (420 Ma), predates the D magnetization, found in the granitic phase (380 Ma). The C and D magnetizations yield poles at 068°E, 15°S and 125°E, 63°S respectively.

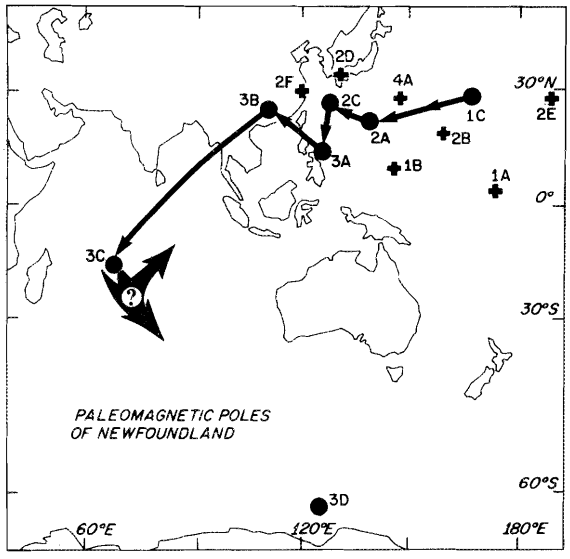


FIG. 12. Paleomagnetic poles from Newfoundland for the Lower Paleozoic period. The trend indicated by the reliable poles is a westward movement from Cambrian to Silurian. The post-Silurian period is yet to be defined. Full dots represent poles obtained after standard paleomagnetic technique. Crosses are poles obtained from NRM measurements or from abstract.

Conclusions

Newfoundland is an important area for the geological study of the Appalachians, but until recently few paleomagnetic data of reliable quality have been available. With the new data presented here it might be appropriate to attempt a correlation of paleomagnetic data of Newfoundland. These data could be of some help toward the understanding of the lower Paleozoic tectonic history of the region. Thus, a brief review of available data for Newfoundland from Cambrian to Devonian is presented (Table 7; Fig. 12). It should be stated that the lower Paleozoic results have not, in general, been supported by corroborative radiometric results; the age control is then open to discussion. Basically, from Cambrian to Silurian there is a linear trend.

From Lower Silurian (3C) to Devonian (4A) there is no data available, except for the Mount Peyton late phase intrusion (3D). This pole is much to the south of 3C and 4A. If this really represents the pole location at about 380 Ma, then a rapid apparent polar wander occurred during the Devonian relative to Newfoundland. This last pole (3D) might also be the first paleomagnetic evidence of an ice age as claimed by McCann and Kennedy (1974) for Newfoundland (paleolatitude = 76°), although the timing of the event seems offset.

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